Name – Meenakshi

Batch id - **7943**

Jenkins with a master-slave architecture using the Jenkins EC2 plugin and implementing RBAC (Role-Based Access Control) access management

Prerequisites:

1. You need an AWS account with appropriate access rights to create EC2 instances.

2. Install Jenkins on a EC2 server.

Task 1: Install Required Plugins

1. Open Jenkins and navigate to "Manage Jenkins" > "Manage Plugins."
2. Install the following plugins if not already installed:
   - EC2 Plugin
   - Role-based Authorization Strategy Plugin (for RBAC)

Task 2: Configure EC2 Plugin

1. Go to "Manage Jenkins" > "Manage node and clouds"
2. Click on "Add a new cloud" > "Amazon EC2."
3. Configure your AWS credentials via IAM role and region.
4. Set up your preferred AMI, instance type, and other EC2 configurations.
5. Test the connection to your AWS account to ensure the plugin can interact with EC2.

Task 3: Set up Jenkins RBAC

Configure role based access control in global security and implement RBAC.

1. Go to "Manage Jenkins" > "Manage and Assign Roles."
2. Click on "Manage Roles" > "Manage Roles."
3. Click on "Create new roles."
4. Create roles based on various requirements (e.g., admin, developer, tester).
5. Assign specific permissions to each role (e.g., job creation, job configuration, global configuration, etc.).

Task 4: Assign Users to Roles

1. Go to "Manage Jenkins" > "Manage and Assign Roles."
2. Click on "Assign Roles."
3. Assign each user to appropriate roles based on their responsibilities.

Task 5: Test the Setup

1. Create a new Jenkins job on the master.
2. In the job configuration, specify which label the job should run on (the label you set for your slave node).
3. Build and run the job to ensure it runs on the EC2 slave node.

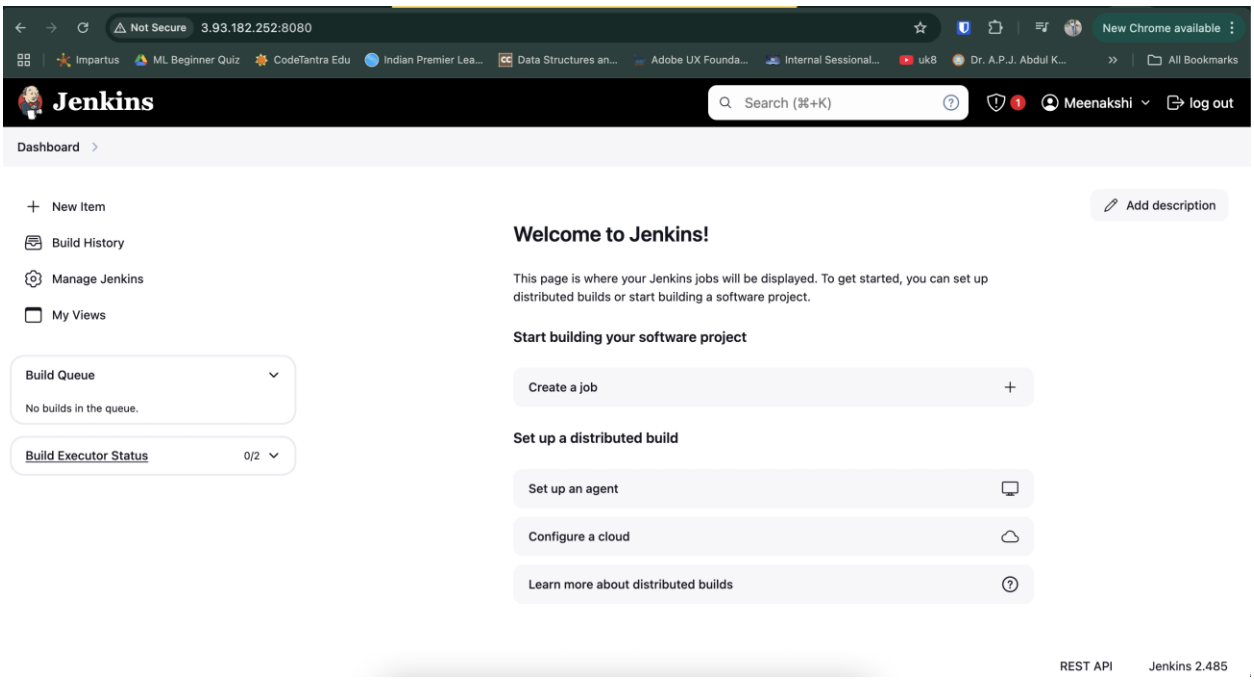
Task 6: Verify RBAC

1. Login to Jenkins with different user accounts.
2. Test whether each user can access and perform actions based on their assigned roles.

Keep in mind that working with AWS resources involves real costs. Make sure to terminate any EC2 instances that you create during this practice project to avoid unexpected charges.

Task 1: Install Required Plugins

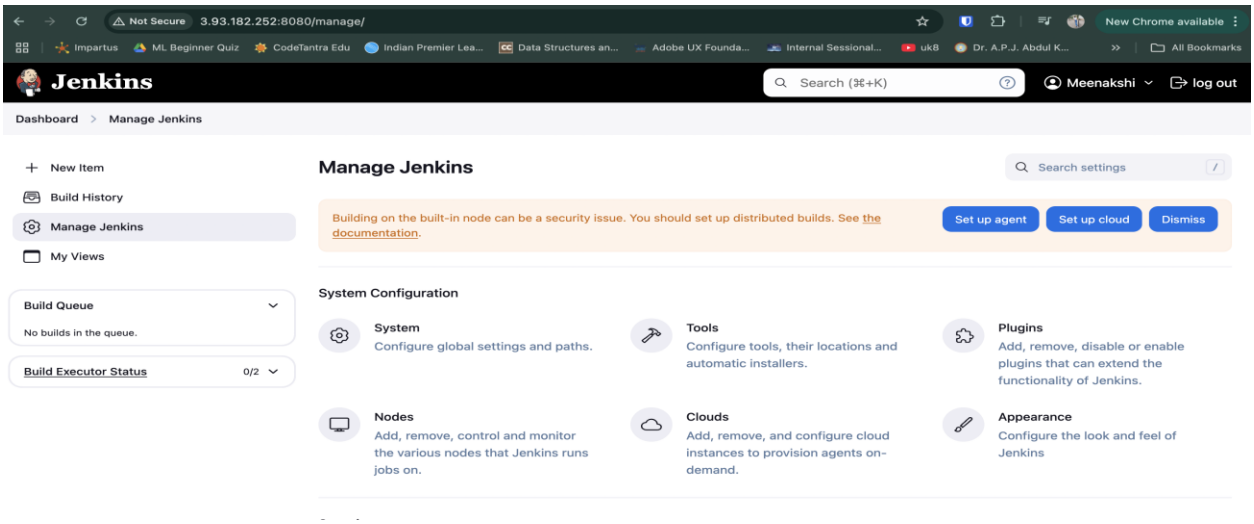
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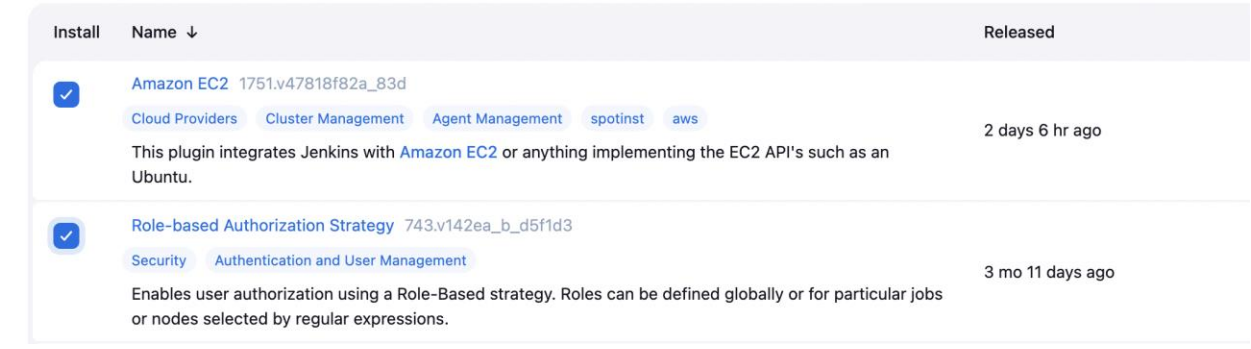


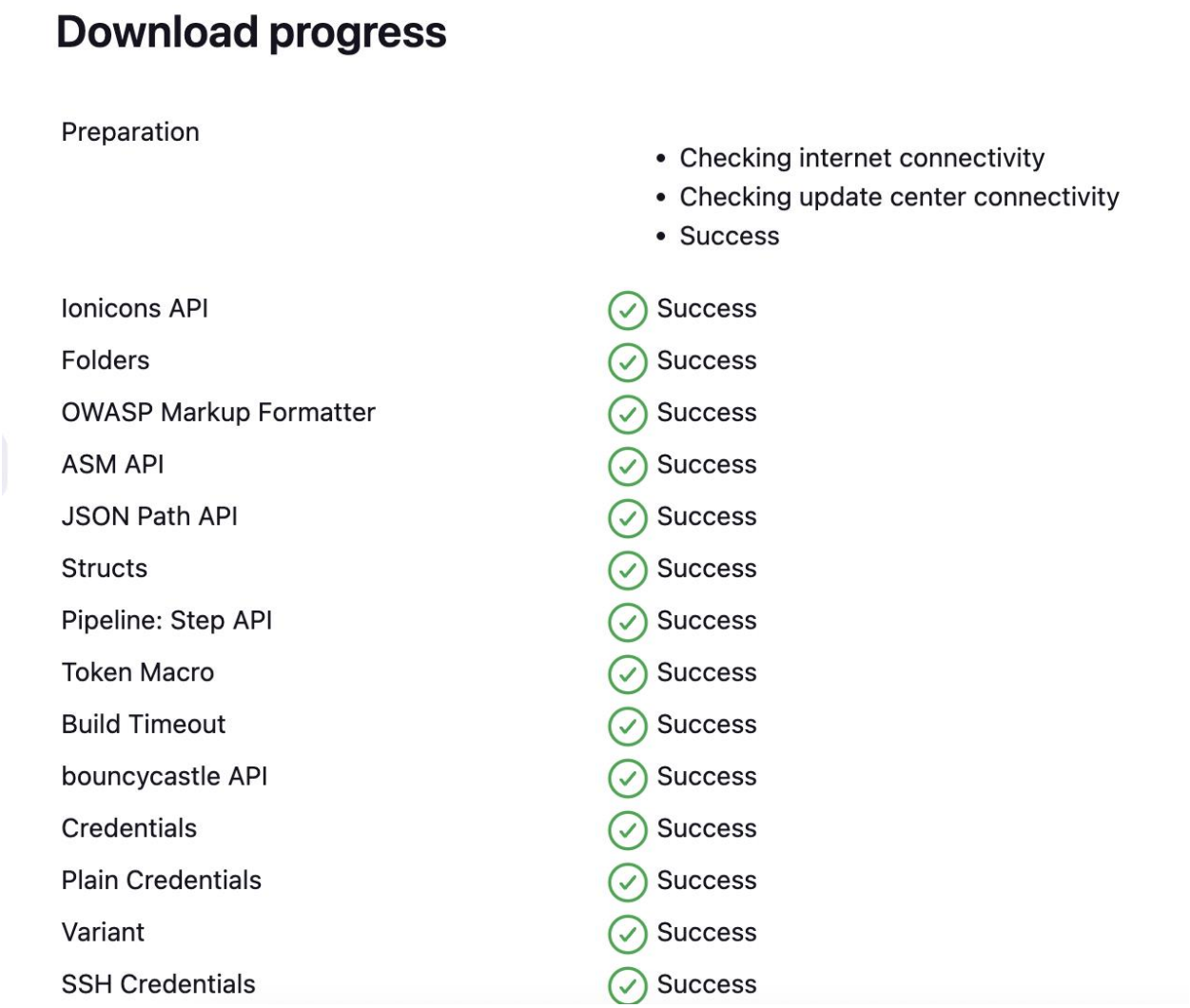
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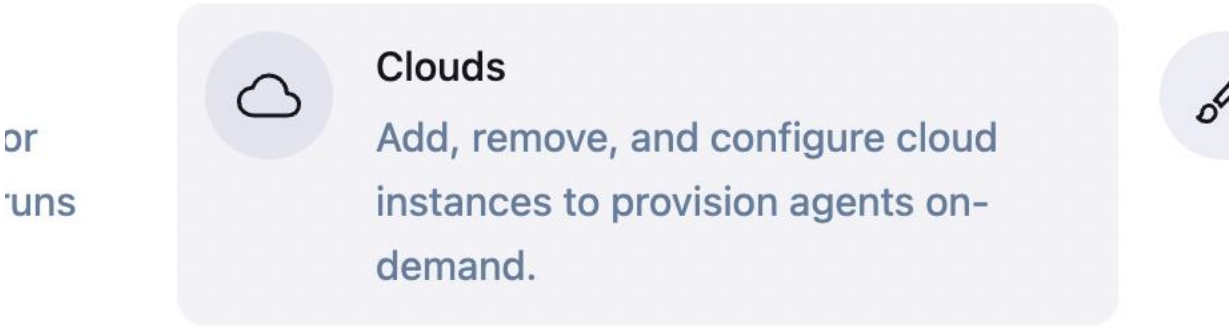






Task 2: Configure EC2 Plugin

1. Go to "Manage Jenkins" > "Manage node and clouds"

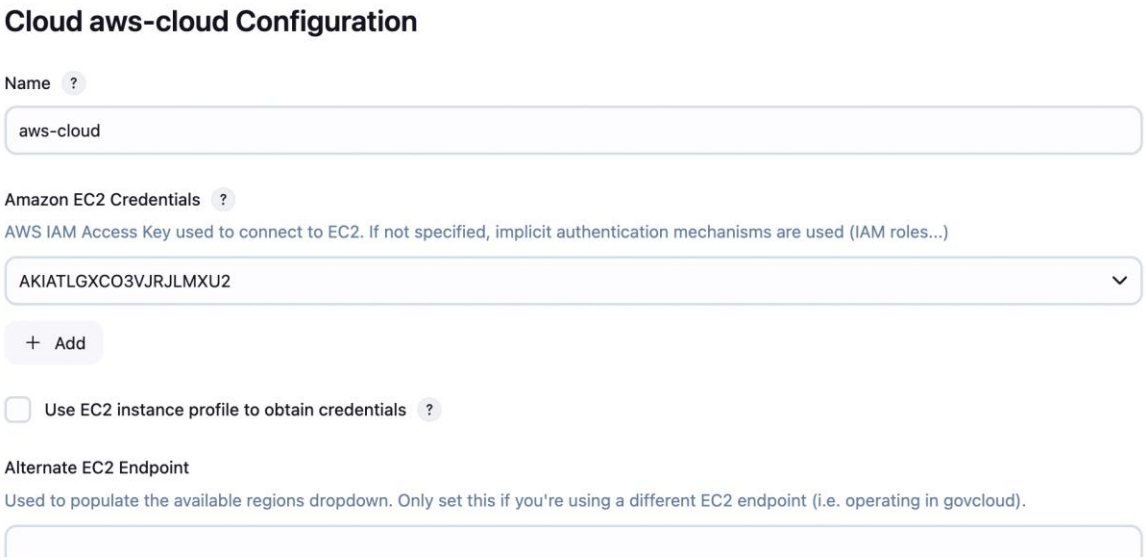


2. Click on "Add a new cloud" > "Amazon EC2."



3. Configure your AWS credentials via IAM role and region

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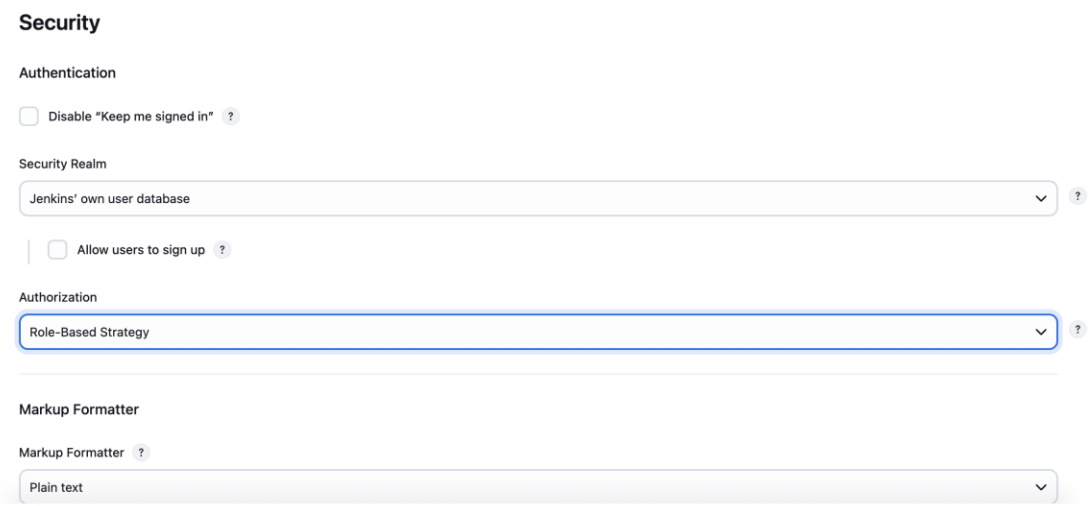
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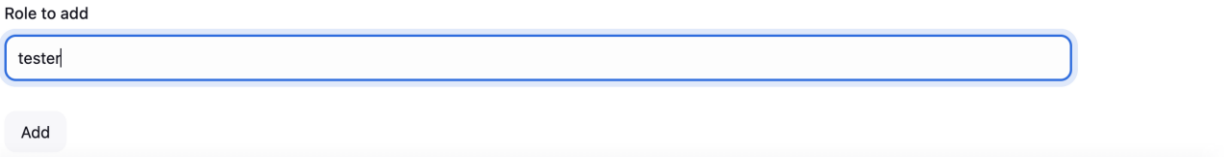
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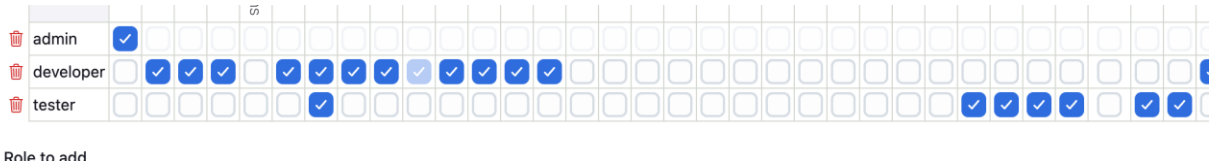
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3. Click on "Create new roles."



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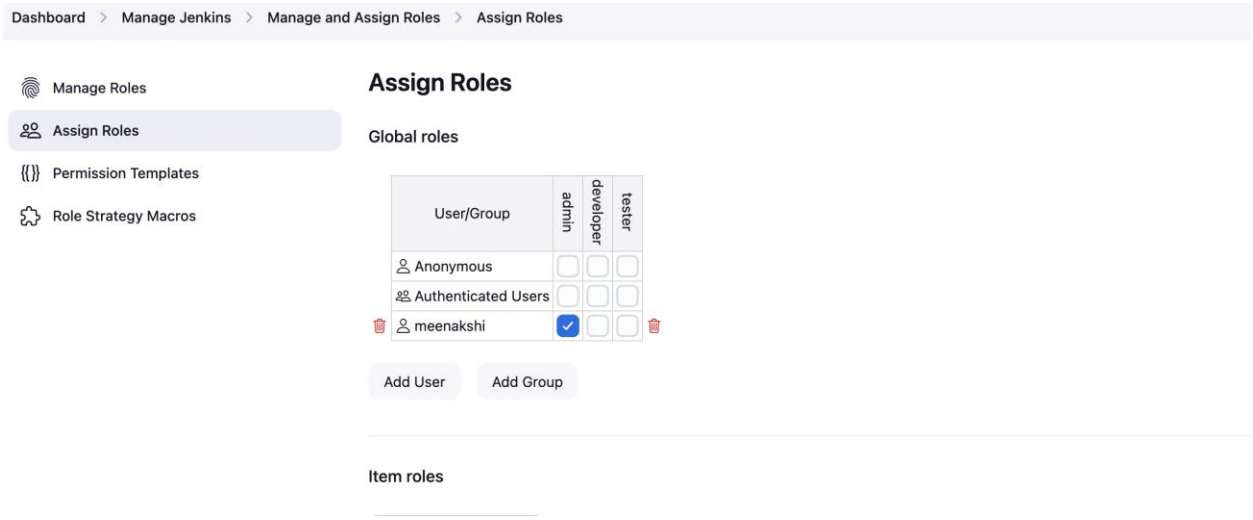


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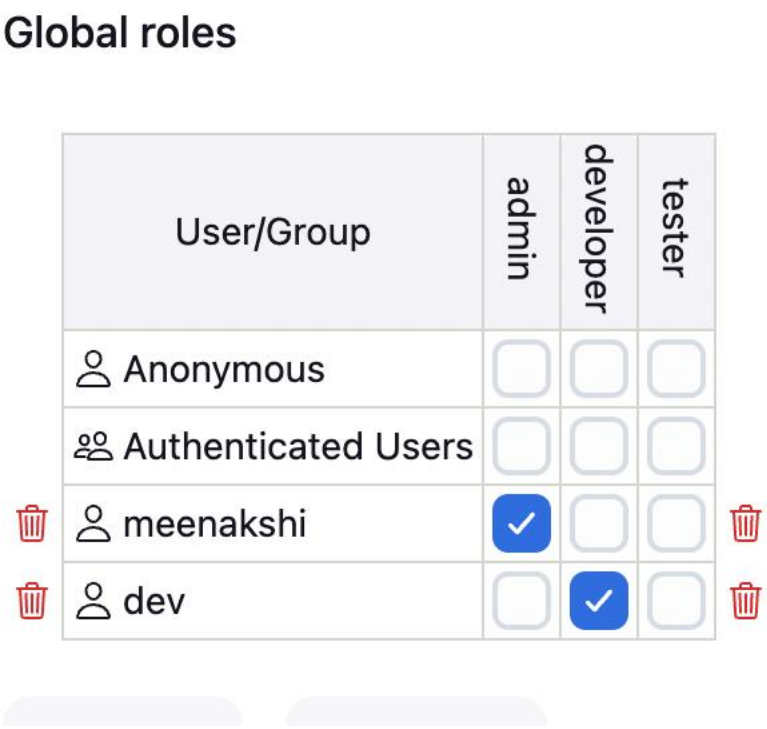
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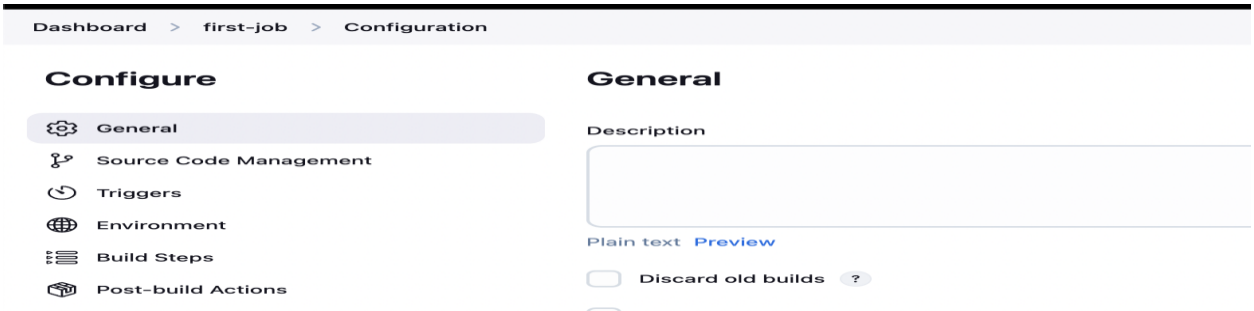


3. Assign each user to appropriate roles based on their responsibilities.

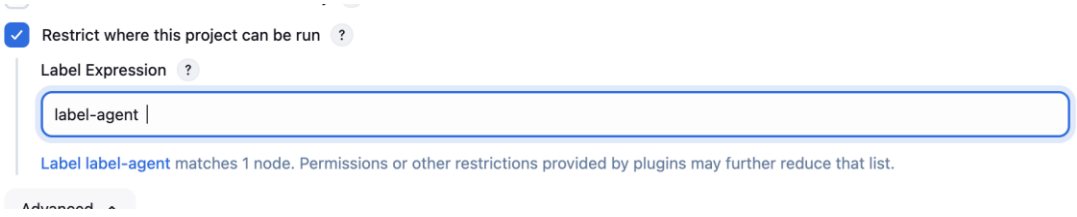


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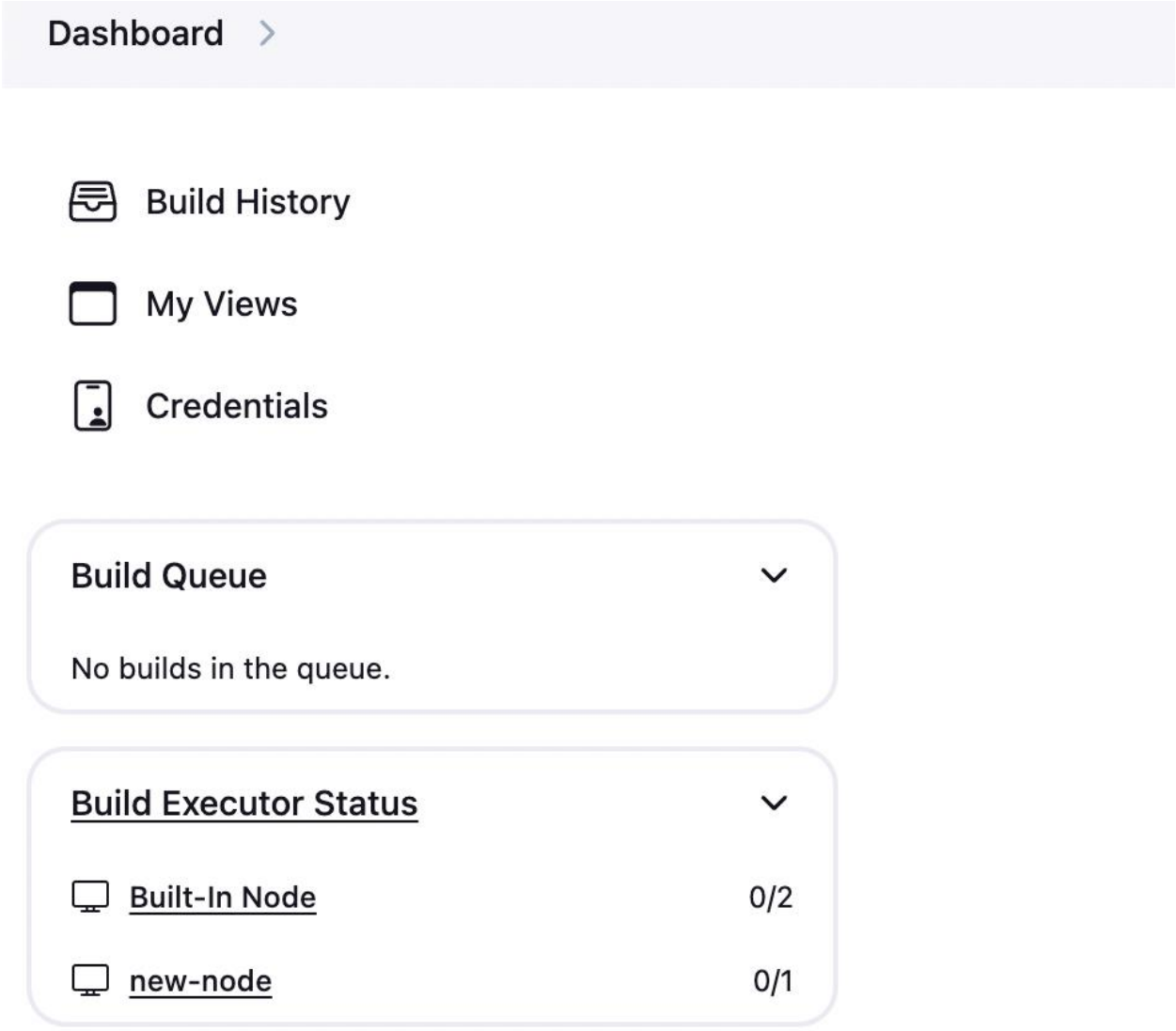


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Ans - terminated